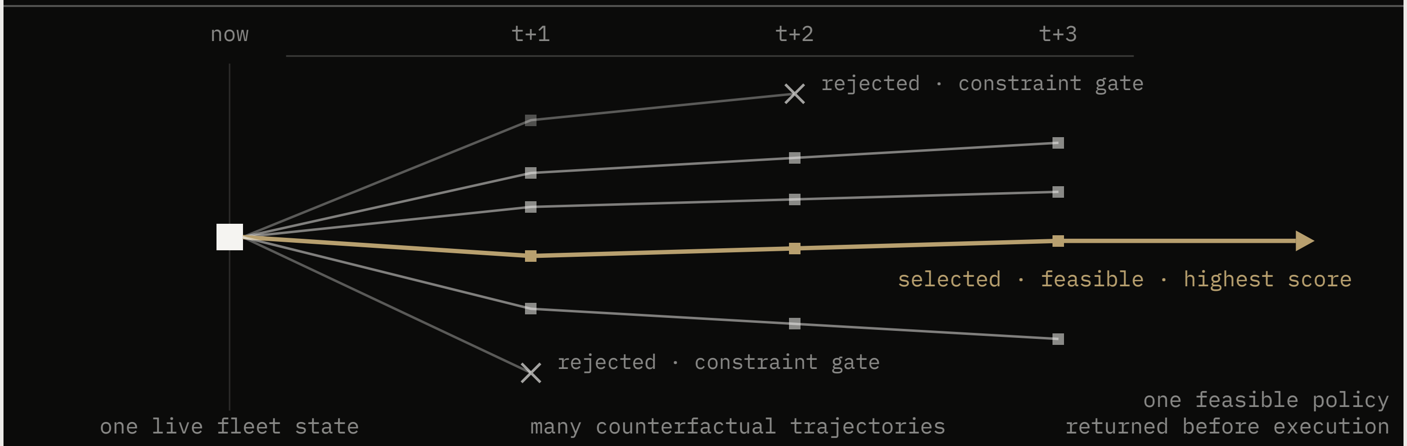
Choose the best fleet trajectory, not just the best next action.

ONE LIVE FLEET STATE · MANY COUNTERFACTUAL FUTURES · ONE SAFE TRAJECTORY SELECTED BEFORE EXECUTION

FIG.01 COUNTERFACTUAL TRAJECTORY SEARCH — SIMULATED · READ-ONLY



CANDIDATE EVALUATION NEVER WRITES TO THE LIVE FLEET — METADATA ONLY

Kubernetes, Slurm, serving schedulers, autoscalers, and fleet-management systems already own specialized control and execution decisions. Aurelius adds a pre-execution planning layer above them.

Before a decision is committed, Aurelius forks the current fleet state into competing simulated trajectories. Each trajectory applies a different coupled policy across the controls implemented for that workload, including batching, routing, placement, capacity, prewarming, admission, migration, precision, speculative decoding, and GPU clock policy.

The fleet state is then advanced through the planning horizon. Each trajectory changes queue progression, capacity use, placement and topology pressure, replica warm state, SLA outcomes, and modeled infrastructure cost.

Trajectories that violate configured SLA, capacity, power, residency, placement, or policy constraints are rejected before ranking. The highest-scoring feasible policy is returned to the existing control plane, which remains authoritative and executes through its normal interfaces.

8.24×

mean SLA-safe goodput per modeled infrastructure dollar versus a constructed production-informed replay baseline

84.5%	fewer GPU-hours under identical input request sets
1,549,249	requests across three selected uncapped high-load windows
3 of 3	windows with lower SLA violation rates

Simulated, load-dependent evidence from selected public-trace windows. Not a production result and not an estimate of performance against a particular fleet or internal scheduler.

The optimizer cannot choose a future it never represents.

The uncapped ablation reran the exact harness, windows, and load behind the 8.24× headline and varied one thing: the search strategy available to the planner. The production anchors reproduced the published values exactly, in all three markets.

FIG.02 SEARCH-STRATEGY ABLATION · UNCAPPED REPLAY — SINGLE VARIABLE: THE SEARCH

Strategy		Ratio	Change
Constructed baseline		1.00×	
GPU clock only · single surface replay exceeds the harness budget	DOES NOT COMPLETE	·	
Fixed 24-policy grid		3.62×	+261.9%
Regime-specific candidate set · no coupled search		3.62×	+261.9%
Exhaustive search of the fixed grid		4.84×	+383.6%
Bounded coupled search · generated set		4.85×	+385.4%
Hierarchical coupled-policy search		8.29×	+728.6%

SLA violation rates fall down the ladder: 0.038 to 0.046 at the baseline; 0.0073 to 0.011 for the grid arms; 0.0022 to 0.0026 for the hierarchical search, the lowest of any arm.

IDENTICAL INPUTS, WINDOWS, AND LOAD IN EVERY ARM · 1,549,249 REQUESTS — RERUN · 07.2026

A clock-only, single-surface search cannot even be scored at this load: the policy it selects degrades the simulated fleet until the replay itself exceeds the harness budget, in all three markets.

Setting the posture jointly is worth roughly 3.6× on its own: a fixed 24-policy grid and the regime-specific candidate set, scored without coupled search, select the same policy at this load.

Searching the grid harder is saturated: exhaustive enumeration of the fixed grid reaches 4.84×, and the bounded coupled search matches it at 4.85× while evaluating far fewer candidates.

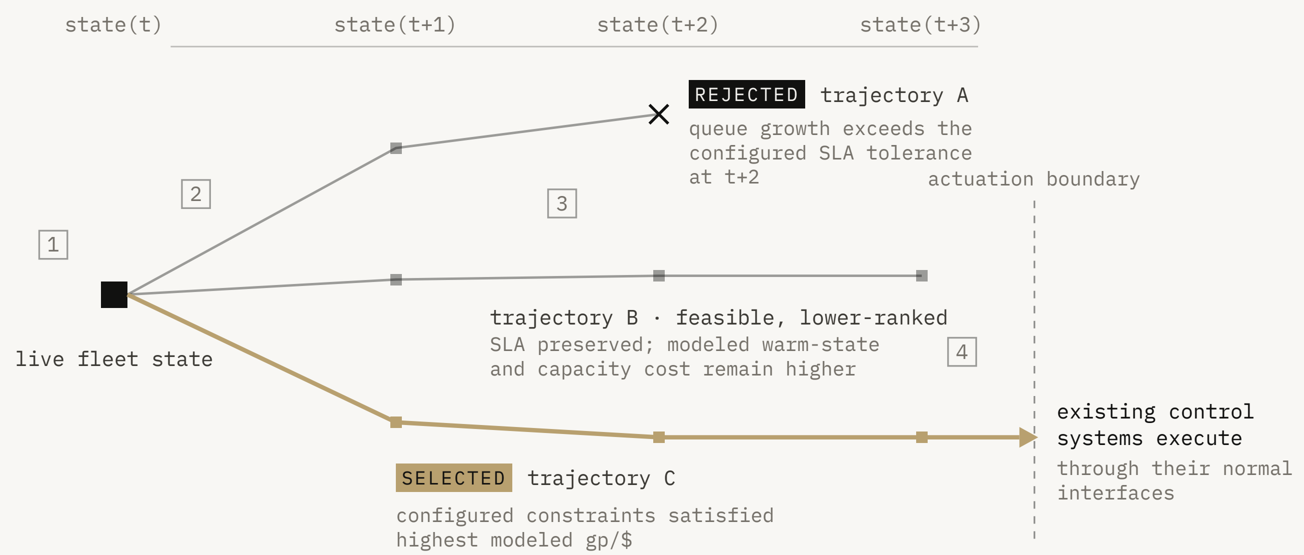
The hierarchical coupled-policy search reaches 8.29× with the lowest SLA violation rates of any arm. The near-doubling beyond every grid-based strategy is value in coupled futures no fixed grid represented. The forecast, simulator, objective, cost model, and constraint gates were unchanged in every arm.

The value was not hidden in a better score. It was hidden in a future the previous search never considered.

Uncapped search-strategy ablation, 07.2026: an independent rerun of the exact harness, windows, and load behind the headline benchmark. Production anchors reproduced published values exactly; the hierarchical arm reproduced the published headline within 1.2 percent per market (8.29× here against the published 8.24× mean). Ratios are unweighted means across the three markets. The frozen V1 audit separately measured zero search regret where exhaustive enumeration was tractable, at approximately 40 candidate evaluations per decision. Neither validates the simulator or establishes production savings.

One fleet. Many futures. Simulated before execution.

FIG.03 STATE-TRANSITION SCHEMATIC · ONE CONTROL PERIOD GATE → RANK → RETURN



EACH CANDIDATE RUNS AGAINST ISOLATED SIMULATED STATE NEVER WRITES TO THE LIVE FLEET

Live fleet state: queue · capacity · placement · topology · replica state · configured constraints.

01 **Mirror the fleet.**

A persistent state-transition model carries queue state, capacity, placement and locality, replica warm and cold state, migrations in flight, configured constraints, and the modeled cost basis across control periods.

02 **Generate coupled policies.**

A regime-specific generator proposes a bounded set of policies based on the workload, operating conditions, implemented controls, and configured constraints. Previously validated candidates are retained as controls.

03 **Advance each trajectory.**

Every candidate is simulated independently over the planning horizon. Forecasts use only information available at the decision timestamp. Future arrivals, future prices, and future outcomes are not supplied to the planner.

04 **Reject infeasible trajectories and return the winner.**

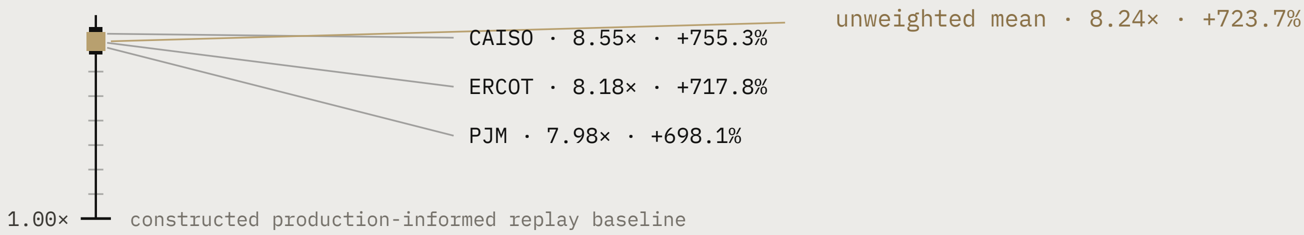
Hard constraints reject infeasible trajectories before economic ranking. The highest-scoring feasible policy is returned as a recommendation. When forecast confidence falls below the configured threshold, Aurelius returns the configured deterministic fallback policy.

Aurelius sits above Kubernetes, Slurm, serving schedulers, routing systems, autoscalers, and fleet-management systems. It evaluates how their decisions interact across a shared planning horizon. It does not replace them, write cluster state, or bypass operator-configured constraints.

The proposed historical replay requires scheduler and fleet metadata. It does not require customer identity, request payloads, model outputs, or proprietary weights.

A signal too large to ignore and too simulated to trust without operator data.

FIG.04 UNCAPPED REPLAY · RATIO TO THE CONSTRUCTED BASELINE THREE MARKET WINDOWS



1,549,249 total input requests · 84.5% fewer total GPU-hours · lower SLA violation rates in all three windows · both policies received the identical input request sets.

The comparison policy is a deterministic public-replay baseline assembled from established serving and cluster-control mechanisms, including continuous batching, deadline-aware ordering and admission, prefix-cache-aware routing, topology-aware placement, and backlog-driven capacity control. It is stronger than the naive SLA-aware reference, but it is not an operator's internal scheduler and does not reproduce a particular fleet's adaptive logic, internal signals, or cost model.

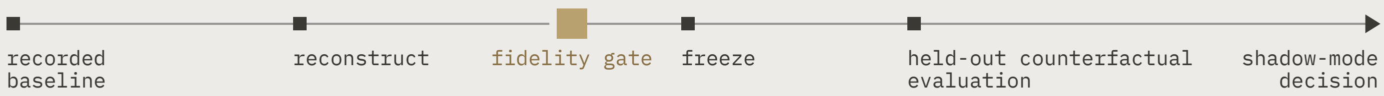
WHAT THIS ESTABLISHES

Under the selected uncapped high-load replay conditions, coupled predictive planning found feasible trajectories with materially better modeled economics and lower SLA violation rates than the constructed baseline.

WHAT THIS DOES NOT ESTABLISH

Expected performance against a particular fleet, internal scheduler, workload distribution, or operator cost model. The result is simulated and load-dependent. Its magnitude is materially affected by the benchmark's batching, precision, service-time, and modeled cost assumptions.

Test one bounded historical window.



INPUT	Scheduler and fleet metadata from one representative workload, cluster, or bounded historical window. No customer identity, request payloads, model outputs, proprietary weights, production actuation, or unrestricted infrastructure access.
PROCESS	First reconstruct the recorded control decisions and verify that the simulator reproduces recorded queue, execution, resource-use, and SLA outcomes within agreed tolerances. Counterfactual improvements are scored only if that baseline-reconstruction fidelity gate passes. Then freeze the calibrated simulator and evaluate Aurelius on a held-out historical window using only information available at each decision timestamp.
OUTPUT	Matched SLA, GPU-hour, utilization, completed-work, and agreed economic comparisons; simulator-fidelity results; sensitivity analysis; attribution by active policy dimension; failure cases and negative findings; and a recommendation on whether recommendation-only shadow mode is warranted.

REQUESTED NEXT STEP

A short scoping session with one engineering owner to select a bounded historical window, validate the minimum metadata schema, and agree on replay acceptance criteria.

If Aurelius cannot reproduce baseline behavior and improve the pre-registered primary metric on held-out telemetry, the evaluation stops. If it can, the next step is recommendation-only shadow mode.

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Complete methodology, references, benchmark tables, limitations, and mechanism audits are in the supporting full technical report.